	Baseline			UNO			UNO Speedup	
Task	CPU	GPU	Speedup	CPU	GPU	Speedup	CPU	GPU
mnistAdd	230.56	107.91	2.14×	55.07	**10.60**	5.19×	4.19×	10.18×
member3	18.32	10.60	1.73×	8.45	**3.48**	2.43×	2.17×	3.05×
member5	818.21	866.51	0.94×	19.47	**4.12**	4.73×	42.02×	210.28×
add2x2	95.73	32.93	2.91×	8.51	**4.37**	1.95×	11.25×	7.53×
card_sum_2	1027.35	170.93	6.01×	950.49	**34.22**	27.78×	1.08×	5.00×
card_sum_3	1623.92	294.29	5.52×	1359.05	**45.97**	29.57×	1.19×	6.40×
card_sum_4	7483.29	4630.94	1.62×	2335.97	**75.26**	31.04×	3.20×	61.53×
card_prodsum_2	1119.02	223.86	5.00×	1016.03	**31.50**	32.26×	1.10×	7.11×
Total	12416.41	6337.98	1.96×	5753.05	**209.51**	27.46×	2.16×	30.25×

Table 1: Elapsed runtime (seconds) for Baseline and UNO on CPU and GPU.

	Baseline			UNO			UNO Speedup	
Task	CPU	GPU	Speedup	CPU	GPU	Speedup	CPU	GPU
mnistAdd	3m51s	1m48s	2.14×	55s	**11s**	5.19×	4.19×	10.18×
member3	18s	11s	1.73×	8s	**3s**	2.43×	2.17×	3.05×
member5	13m38s	14m27s	0.94×	19s	**4s**	4.73×	42.02×	210.28×
add2x2	1m36s	33s	2.91×	9s	**4s**	1.95×	11.25×	7.53×
card_sum_2	17m7s	2m51s	6.01×	15m50s	**34s**	27.78×	1.08×	5.00×
card_sum_3	27m4s	4m54s	5.52×	22m39s	**46s**	29.57×	1.19×	6.40×
card_sum_4	2h4m43s	1h17m11s	1.62×	38m56s	**1m15s**	31.04×	3.20×	61.53×
card_prodsum_2	18m39s	3m44s	5.00×	16m56s	**31s**	32.26×	1.10×	7.11×
Total	3h26m56s	1h45m38s	1.96×	1h35m53s	**3m30s**	27.46×	2.16×	30.25×

Table 2: Elapsed runtime (hours, minutes, seconds) for Baseline and UNO on CPU and GPU.

Notes for both time tables. Within each system, speedup is CPU time divided by GPU time. UNO speedup is Baseline time divided by UNO time on the same device.

	Baseline		UNO	
Task	CPU	GPU	CPU	GPU
mnistAdd	**98.63**	**98.63**	98.60	98.50
member3	95.00	95.36	95.20	**95.52**
member5	92.78	**93.20**	92.91	92.19
add2x2	**96.44**	96.30	96.20	96.22
card_sum_2	97.75	97.55	97.85	**98.40**
card_sum_3	91.97	88.50	92.77	**94.93**
card_sum_4	75.71	73.91	70.33	**83.49**
card_prodsum_2	59.20	**62.35**	56.45	52.15
Mean	88.44	88.22	87.54	**88.93**

Table 3: Latent test accuracy (%) for Baseline and UNO on CPU and GPU. Latent accuracy measures image classification; values are percentages. Mean is the unweighted task mean. Higher accuracy is better.

Hardware:

CPU: Intel Core Ultra 7 265 5.3 GHz (20 cores, 20 logical CPUs)

Memory: 62.25 GiB usable RAM

GPU: NVIDIA RTX PRO 5000 Blackwell, 48935 MiB

Run configuration. All runs use the same seed, 1537081416, read from the machine `/dev/urandom`. They are all run for 3 epochs with batch size 16. The Baseline runs use "`--checkpoint_freq 0`" to avoid writing checkpoints. All runs use stable-model caching, which is enabled by default for Baseline and UNO.